

Question

There are the following six chemical processes:

1. Under alkaline conditions, reduce $\text{Cu}(\text{NH}_3)_4^{2+}$ with $\text{Na}_2\text{S}_2\text{O}_4$ to obtain a colorless complex.
2. Add excess sodium cyanide to a solution containing copper ions, causing the blue color of the solution to disappear, while releasing a highly toxic gas.
3. Boil copper(I) chloride in concentrated NaOH solution, where the copper element is completely converted into a red solid.
4. The hydrolysis disproportionation of IF_3 yields three common inorganic substances.
5. Dissolve elemental Pb in sodium hydroxide and write the ionic equation for the reaction.
6. Treat $\text{Mn}(\text{IO}_3)_2$ with bromine trifluoride at 800 K to prepare a fluoride, with another fluoride also formed in the products.

Write and balance the equations for the above processes, simplifying the coefficients to the smallest integer ratios. For chemical processes in aqueous solutions, use ionic equations. The following statements are correct:

A. Chemical Process 1 produces SO_2 gas.

B. Chemical process 2 produces complex ions of copper, where copper has a coordination number of 2.

C. The gas produced by chemical process 2 is colorless and odorless.

D. After balancing the chemical equation for Process 3, the sum of coefficients on the left and right sides is unequal.

- E.** One of the products in Chemical Process 4 exhibits reducing properties and can undergo a displacement reaction with bromine.
- F.** Chemical process 5 produces a lead complex in which lead is in the +4 valence state.
- G.** One of the products of Chemical Process 6 has a square pyramidal structure, and the sum of the product coefficients on the right side of the equation is 49.
- H.** Chemical Process 6 produces a pale yellow gas.
- I.** In the above six processes, three involve gaseous products.
- J.** Among the six equations above, there are two reactions where the sum of coefficients on both sides is equal.
- K.** The sum of product coefficients in the chemical process 2 equations is the smallest among the 6 equations.
- L.** None of the above options is correct

Answer

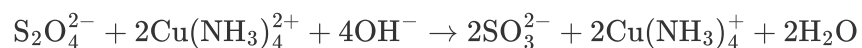
Correct Answer: **G**

Detailed Explanation

Chemical Process 1:

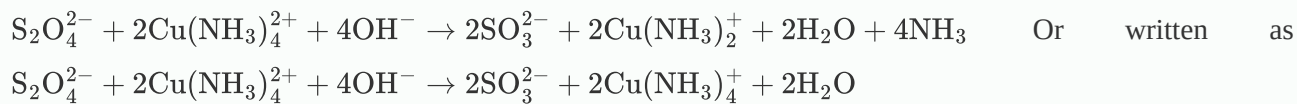


Or written as



CHECKPOINT

1 PTS



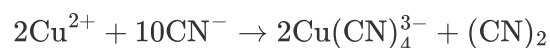
Exists as SO_3^{2-} under alkaline conditions, A is wrong

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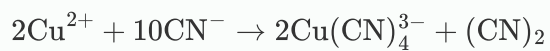
Exists as SO_3^{2-} under alkaline conditions, A is wrong

Chemical Process 2:



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Excess sodium cyanide, Cu should be 4-coordinated, cyanogen is colorless with a bitter almond smell, BC is wrong

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Excess sodium cyanide, Cu should be 4-coordinated, cyanogen is colorless with a bitter almond smell, BC is wrong

Chemical Process 3:



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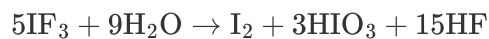
The sum of coefficients on both sides are equal, D is wrong

CHECKPOINT

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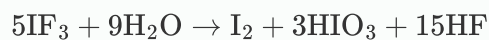
The sum of coefficients on both sides are equal, D is wrong

Chemical Process 4:



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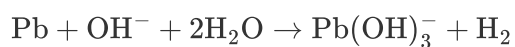
The product is iodine element instead of hydroiodic acid, hydroiodic acid has reducing properties and can undergo disproportionation reaction with iodic acid, E is wrong

CHECKPOINT

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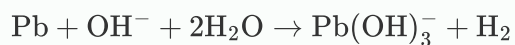
The product is iodine element instead of hydroiodic acid, E is wrong

Chemical Process 5:



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Lead is in the +2 valence state, F is wrong

CHECKPOINT

1 PTS

Lead is in the +2 valence state, F is wrong

Chemical Process 6:



CHECKPOINT

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The product is IF_5 , a tetragonal pyramidal structure, the sum of the coefficients of the products on the right side is 49, +7 I can oxidize Mn^{2+} ions to permanganate, so the product cannot be IF_7 , G is correct, at this reaction temperature, the gaseous products are only Br_2 and O_2 , which are reddish-brown and colorless, respectively, H is wrong

CHECKPOINT

1 PTS

The product is IF_5 , a tetragonal pyramidal structure, the sum of the coefficients of the products on the right side is 49, G is correct

CHECKPOINT

1 PTS

The gaseous products are only Br_2 and O_2 , which are reddish-brown and colorless, respectively, H is wrong

Among the 6 processes, 1, 2, 5, and 6 all produce gas, I is wrong; only in 3 the sum of coefficients on both sides are equal, J is wrong; the process with the smallest sum of product coefficients is process 5, K is wrong

CHECKPOINT

1 PTS

Among the 6 processes, 1, 2, 5, and 6 all produce gas, I is wrong

CHECKPOINT

1 PTS

Only in 3 the sum of coefficients on both sides are equal, J is wrong

CHECKPOINT

1 PTS

The process with the smallest sum of product coefficients is process 5, K is wrong